

Student Name.

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Gene Locus Symbol.

METTL13

Part I. Basic literature search and databases.**1. Known names**

METTL13, CGI-01, FEAT

Bonus question: indicate which of these names are ambiguous.

None of the aliases are ambiguous, as none of the other aliases could refer to any other gene.

2. Basic information about the gene locus.

METTL stands for methyltransferase-like, indicating METTL13 is a protein involved in RNA methylation and affects mRNA translation. Specifically, METTL13 acts as a “eEF1A methyltransferase with dual specificity, which is involved in the codon-specific modulation of mRNA translation.” [1]. METTL13 is “the enzyme responsible for methylation of eEF1A at the N terminus and Lys55... [The] loss of METTL13 activity in cells has functional consequences and results in altered translation rate of specific codons.” [1] “[METTL13] is an intracellular protein that potently drives tumorigenesis in vivo. It is only weakly expressed in normal human tissues, including the testis. In contrast, [METTL13] is aberrantly upregulated in most human cancers.” [2] “Significantly higher plasma [METTL13] levels in cancer patients than healthy subjects suggest that [METTL13] is released from the cytoplasm of tumor cells.” [2] “[Studies] suggest that immunogenic [METTL13] protein circulating in the bloodstream provides a resource for applications in early diagnosis and secondary prevention of some cancers”. [2]

References:

1. Jakobsson, M. E., Małecki, J. M., Halabelian, L., Nilges, B. S., Pinto, R., Kudithipudi, S., Munk, S., Davydova, E., Zuhairi, F. R., Arrowsmith, C. H., Jeltsch, A., Leidel, S. A., Olsen, J. V., & Falnes, P. Ø. (2018). The dual methyltransferase METTL13 targets N terminus and Lys55 of eEF1A and modulates codon-specific translation rates. *Nature communications*, 9(1), 3411. <https://doi.org/10.1038/s41467-018-05646-y>
2. Li, Y., Kobayashi, K., Mona, M. M., Satomi, C., Okano, S., Inoue, H., Tani, K., & Takahashi, A. (2016). Immunogenic FEAT protein circulates in the bloodstream of cancer patients. *Journal of translational medicine*, 14(1), 275. <https://doi.org/10.1186/s12967-016-1034-2>

3. Known RNA isoforms.

NM_014955.3, NM_001007239.2, NM_015935.5

4. Longest proteoform.

>NP_015935.5 Homo sapiens methyltransferase 13, eEF1A N-terminus and K55 (METTL13), transcript variant 1, mRNA

MNLLPKSSREFGSVDYWEKFFQQRGKKAFEWYGTYLELCGVLHKYIKPREKVLVIGCGNSELSEQLYDVGYRDIVN
DISEVVIKQMKECNATRRPQMSFLKMDMTQMEFPDASFQVVLDKGTLDVLTDEEEKTLQQVDRMLAEVGRVLQVGG
RYLCISLAQAHILKKAVGHFSREGWMVRVHQVANSQDQVLEAEPQFSLPVFAFIMTKFRPVPGSALQIFELCAQEQR
KPVRLSAERLAEAVQERQQYAWLCSQLRRKARLGSVSLDLCDGDTGEPYTLHVVDSPVTKPSRDNHFIFII PQG
RETEWLFGMDEGRKQLAASAGFRRLITVALHRGQQYESMDHIQAELSARVMELAPAGMPTQQQVPFLSVGGDIGVRT
VQHQDCSPLSGDYVIEDVQGGDKRYFRRLIFLSNRNVVQSEARLLKDVSHKAQKKRKKDRKKQRPADAEDLPAAPGQ
SIDKSYLCCEHHKAMIAGLALLRNPELLLEIPLALLVVGLGGGSLPLFVVDHDFPKSCIDAVEIDPSMLEVATQWFGF
SQSDRMKVHIADGLDYIASLAGGGEARPCYDVIMFDVDSKDPTLGMSCPPPAFVEQSFLQKVKSILTPEGVFILNLV
CRDLGLKDSVLAGLKAVFPLLYVRRIEGEVNEILFCQLHPEQKLATPELLETAQALERTLRKPGRGWDDTYVLSDMK
TVKIV

5. CDS and UTRs of the longest RNA isoform encoding the longest proteoform.

5' UTR: 1...308

CDS: 309...2408

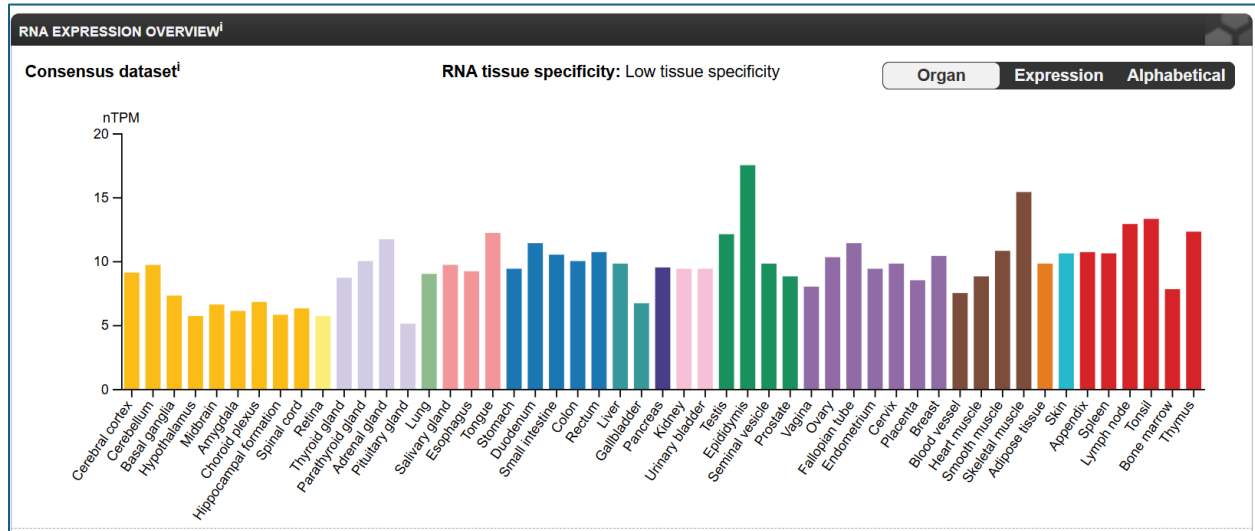
3' UTR: 2409...3368

Bonus question: what is the shortest and what is the longest region in this mRNA. Why are they shorter or longer than the other two?

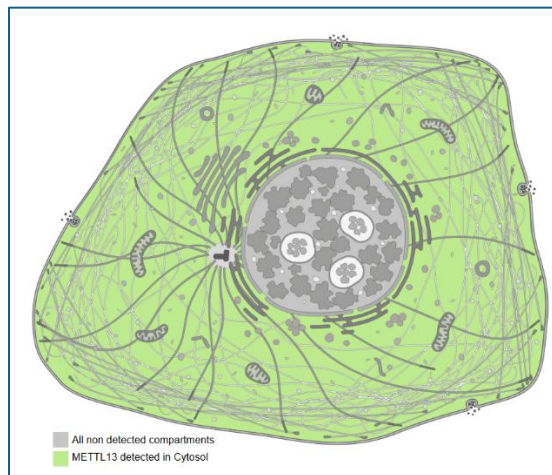
The shortest is the 5'UTR, and the longest is the CD. Typically, the CDs are the longest regions so this is not surprising. The reason why the 3'UTR is longer than the 5'UTR could be because METTL13 undergoes stronger post transcription regulation. This makes sense for a gene implicated in cancer because greater post transcription regulation could add complexity to how the gene controls growth and cell division.

6. Protein localization.

METTL13 is found in all tissues but is most abundantly found in the epididymis and skeletal muscle according to the consensus dataset.



METTL13 is localized to the cytosol.



Bonus question: try to explain the protein localisation based on its function.

METTL13 interacts with eEF1A which is found on ribosomes in the cytosol, affecting translation which occurs exclusively in the cytosol

7. Differential gene expression.

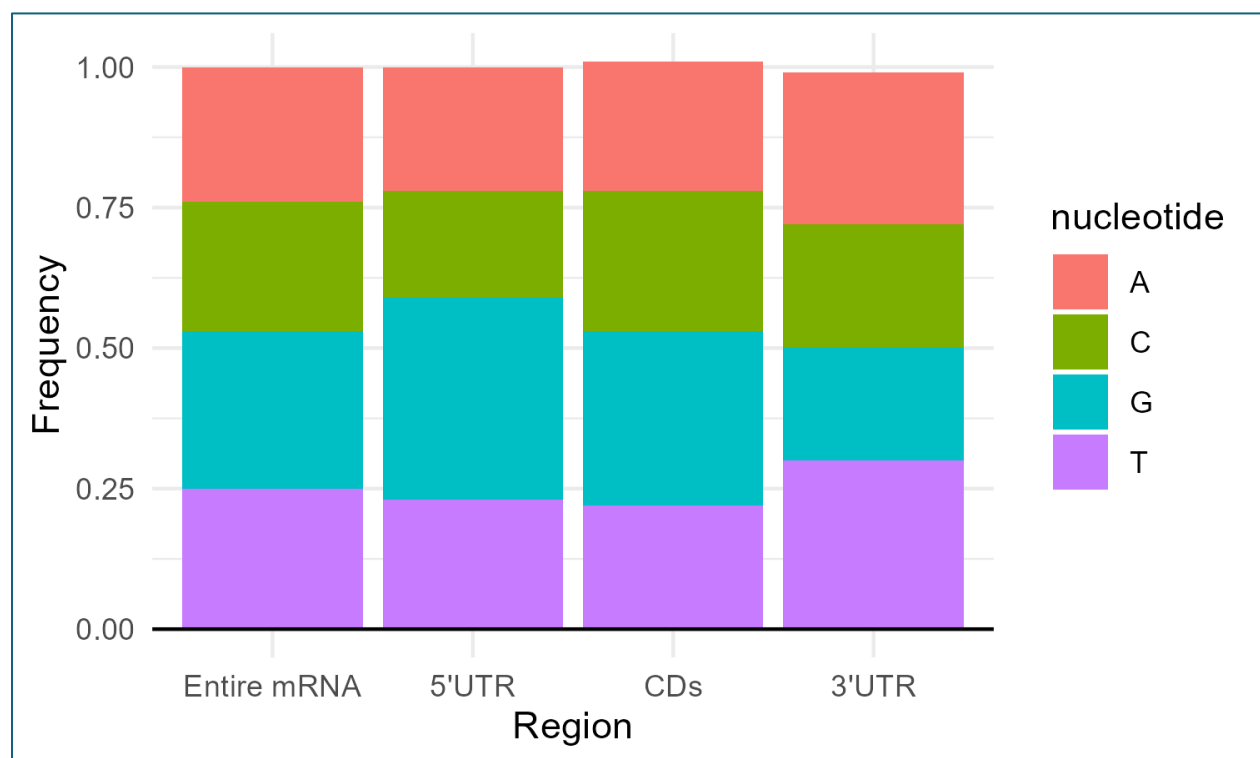
According to EBI Expression Atlas, METTL13 is most upregulated when T cells are stimulated with CD3/CD28 beads (~ 8-fold increase), and most downregulated in those with colorectal carcinoma (~ 4-fold decrease)

Bonus question: try to find a relationship between these changes in gene expression and the protein function.

Part II. Frequency analysis. EMBOSS tools.

1. Nucleotide frequencies.

	Entire mRNA	5'UTR	CDs	3'UTR
A	24%	22%	23%	27%
C	23%	19%	25%	22%
G	28%	36%	31%	20%
T	25%	23%	22%	30%



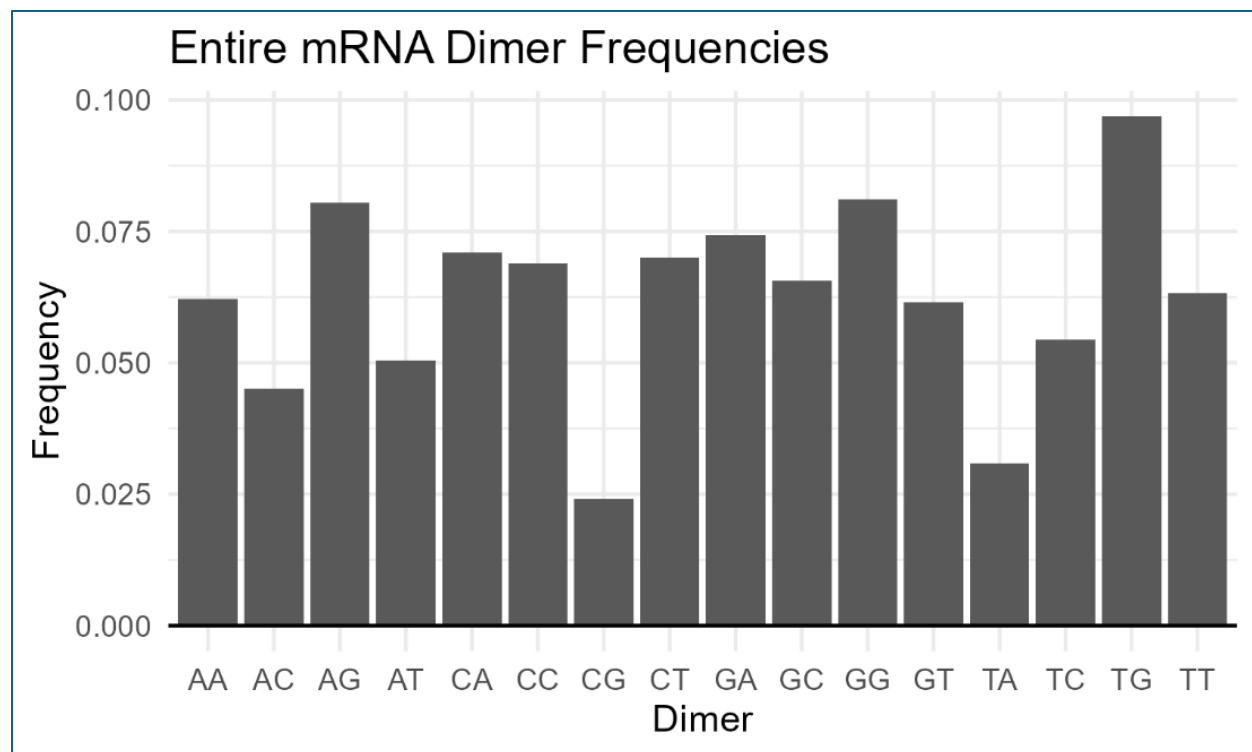
Bonus question: Do these frequencies follow Chargaff rule? Explain why.

No, we see that in these frequencies that while the A-T frequencies are quite similar, the C-G frequencies are quite different, unlike what we'd expect to see if the frequency followed Chargaff's rule.

2. Dinucleotide frequencies.

	Entire mRNA	5'UTR	CDs	3'UTR
AA	6%	8%	5%	8%
AC	5%	3%	4%	5%
AG	8%	7%	8%	7%

AT	5%	4%	5%	6%
CA	7%	6%	7%	7%
CC	7%	4%	8%	6%
CG	2%	5%	3%	.4%
CT	7%	4%	7%	8%
GA	7%	6%	8%	6%
GC	7%	6%	7%	5%
GG	8%	15%	9%	4%
GT	6%	9%	6%	6%
TA	3%	2%	2%	6%
TC	5%	7%	5%	6%
TG	10%	9%	10%	1%
TT	6%	5%	5%	10%



Bonus question: Could you explain the observed differences?

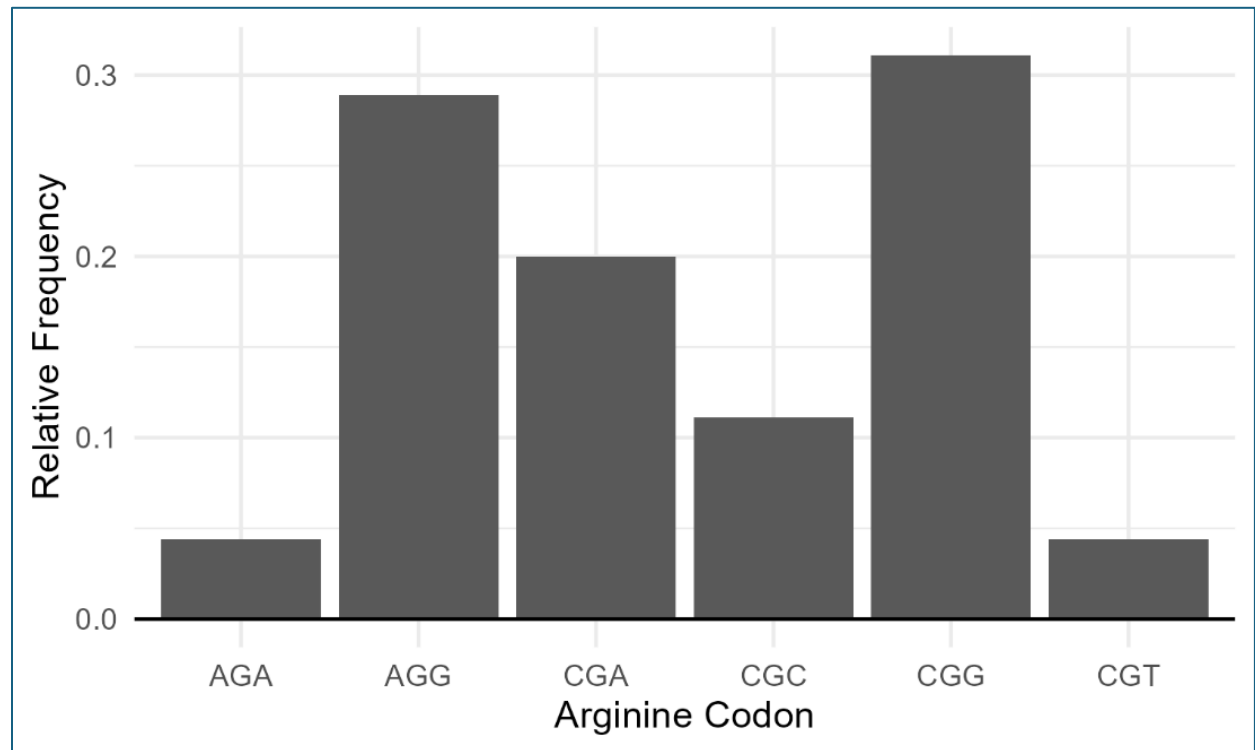
3. Codon usage.

Most frequent: GTG (V) occurs 36 times (5.1%)

Least frequent (besides stop codons): GTA (V) occurs 1 time (0.1%)

Codon	Relative Frequency	Count
AGA	4.4%	2
AGG	28.9%	13

CGA	20%	9
CGC	11.1%	5
CGG	31.1%	14
CGT	4.4%	2

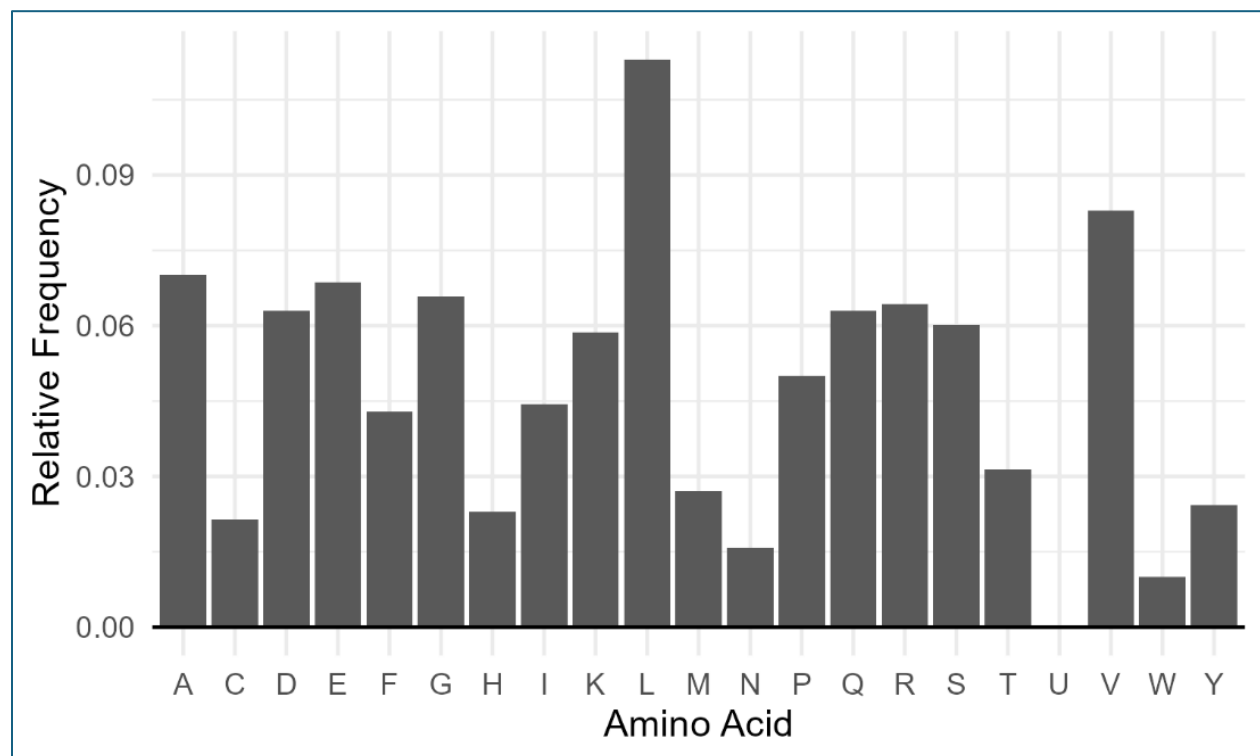


Bonus question: Could you explain these differences?

4. Amino acid frequencies.

Amino Acid	Count	Frequency
A	49	7%
C	15	2%
D	44	6%
E	48	7%
F	30	4%
G	46	7%
H	16	2%
I	31	4%
K	14	6%
L	79	11%
M	19	3%
N	11	2%
P	35	5%
Q	44	6%

R	45	6%
S	42	6%
T	22	3%
U	0	0%
V	58	8%
W	7	1%
Y	17	2%



Bonus question: Could you explain the observed differences?

5. Longest repeat.

There is one twice repeated sequence of 11 nucleotides: CAGTGCTGGCT (1287-1297 & 2183-2193)

6. Longest inverted repeat.

There is one palindrome that is 11 nucleotides long

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113      gtggtccatgc      123
          |||||
1355     caccaggtacg     1345

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7. Chaos Game Representation.

This is a bonus assignment. If you wish to complete it, do it after finishing the entire practical. Obtain the DNA sequence of about 100,000 nucleotides that contains the entire genomic segment of your gene locus. Using EMBOSS chaos generate Chaos Game Representation plot for this sequence. Choose another random genomic sequence roughly of a similar size and make another plot. Now choose a sequence of yeast genome of a similar length and make the third plot. Indicate genomic coordinates for each of the three sequences and include them along with three plots in your report.

Bonus question: *Do you find any similarities between plots? Which plots are more similar to each other? Could you explain the source of this similarity?*

Part III. Sequence similarity. Multiple Sequence Alignment.

1. Orthologs.

The most similar ortholog is from chimpanzee: XP_016788050.2

Alignment score: 1421 bits, % identity: 99.28

The least similar ortholog is from zebrafish: NP_001038234.3

Alignment score: 693 bits, % identity: 53.84

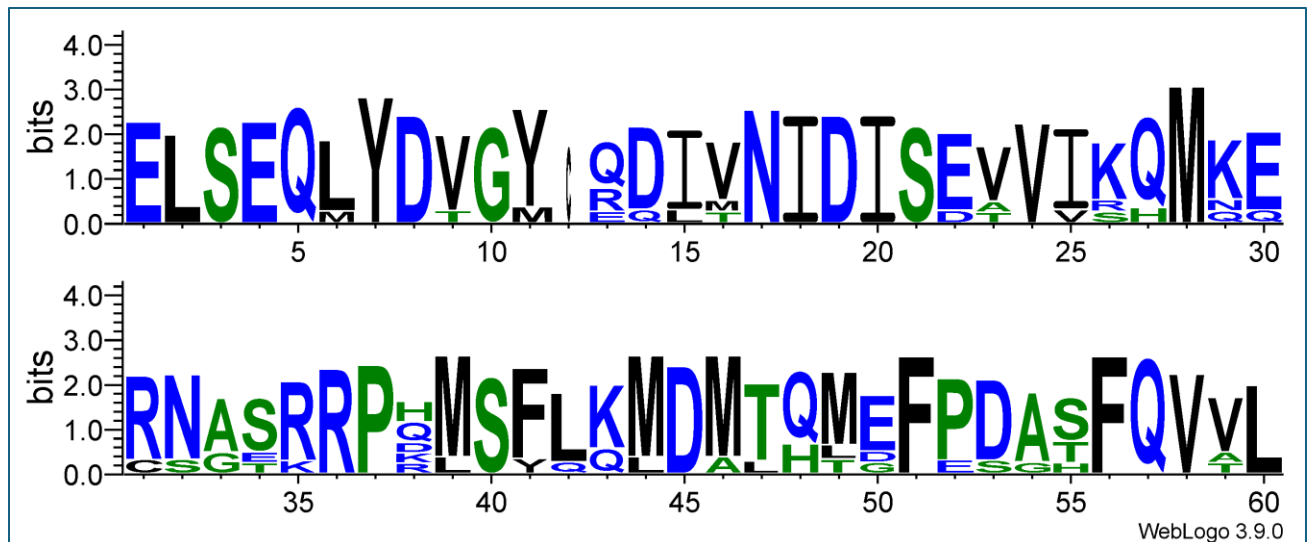
Bonus question: *Why do you think sequences from these particular species are the most and the least similar to their protein homolog?*

The chimpanzee sequence is the closest most likely due to the close evolutionary distance between humans and chimpanzees. On the other hand, the zebrafish is the farthest evolutionary from humans out of the other animals so it makes sense that it is the least similar.

2. Multiple sequence alignment.

NP_001038234.3	ELSEQLYDVGY-RQLTNIDISETVVSHMNQRNAERRPDLSFQQLDATQTGFESGSFQVTL
XP_040560927.1	ELSEQMYDTGMCEDIVNIDISDAVIRQMQERSASKRPKMSYLMQMDMLHMDFPDAHFOVAL
NP_001092555.1	ELSEQLYDVGY-QDIVNIDISEVVIKQMKERNASRRPMSFLKMDMTQMEFPDASFQVVL
XP_016788050.2	ELSEQLYDVGY-RDIVNIDISEVVIKQMKECNATRRPQMSFLKMDMTQMEFPDASFQVVL
XP_025286117.1	ELSEQLYDVGY-QDIVNIDISEVVIKQMKERNASRRPQMSFLKMDMTQMEFPDASFQVVL
NP_001421014.1	ELSEQLYDVGY-QDIMNIDISEVVIKQMKERNASRRPMSFLKMDMTQMEFPDASFQVVL
NP_659126.1	ELSEQLYDVGY-QDIVNIDISEVVIKQMKERNASRRPMSFLKMDMTQLEFPDASFQVVL
	*****:***.* .:: *****:.*: :*: . . :** :*: :*: : * .. ***.*

3. Sequence Logo.



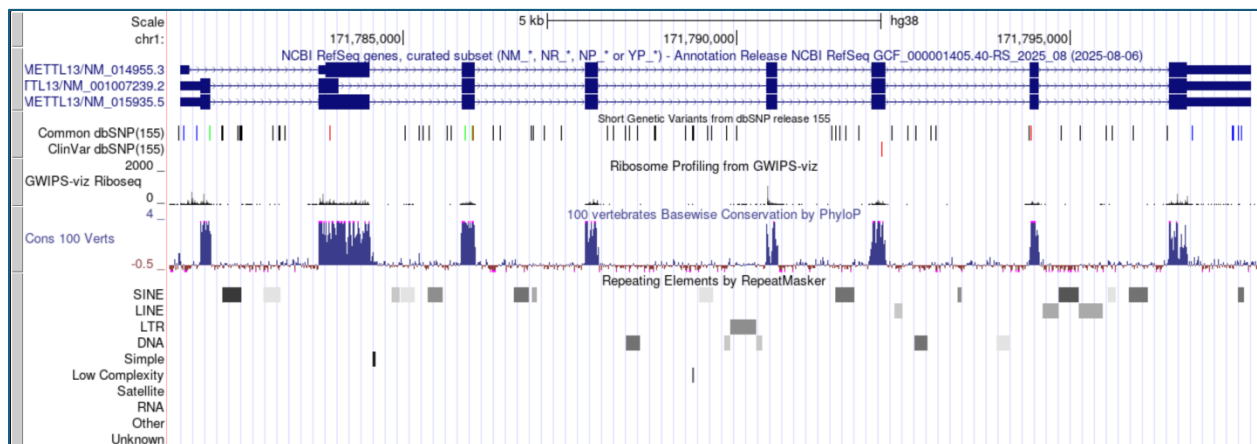
The most conserved regions are the first 5 amino acids (1-5), and 17-21 because they are universal among the various orthologs.

The least conserved region is from 41-50. Not many of the amino acids are fully conserved and most have at least 2 competing amino acids.

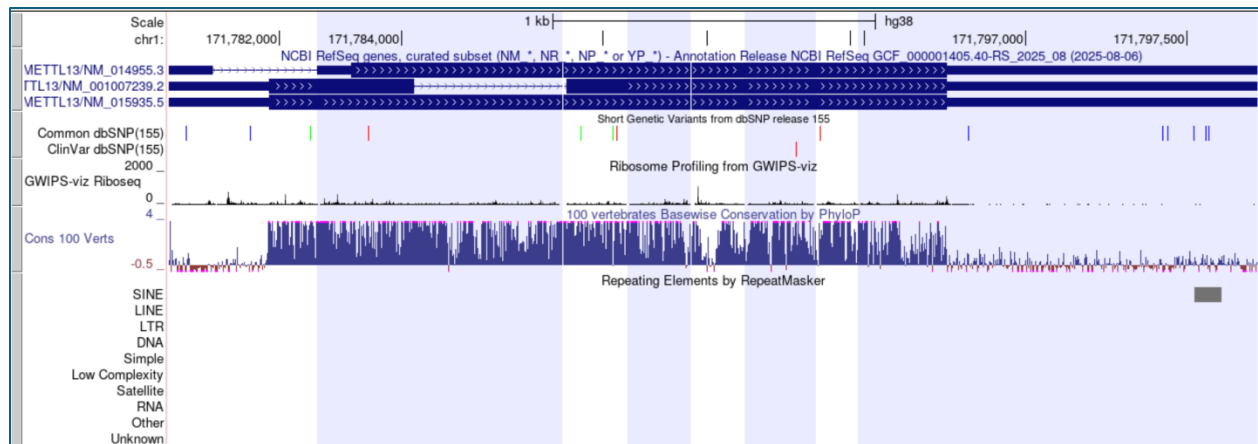
Part IV. Genome browsers.

1. UCSC Genome browser.

METTL13 locus in UCSC Genome Browser:



METTL13 locus in UCSC Genome Browser (multi-region view with most introns removed):



Genomic coordinates: chr1:171,781,660-171,797,717

The number of RNA isoforms: 3

The total number of exons in the mRNA: 8

The number of coding exons in the mRNA: 8

The number of common SNPs in the mRNA: 14

The number of clinically associated SNPs (clinvar) in the mRNA: 1

The number of SINEs and LINEs associated repeats in the locus: 17

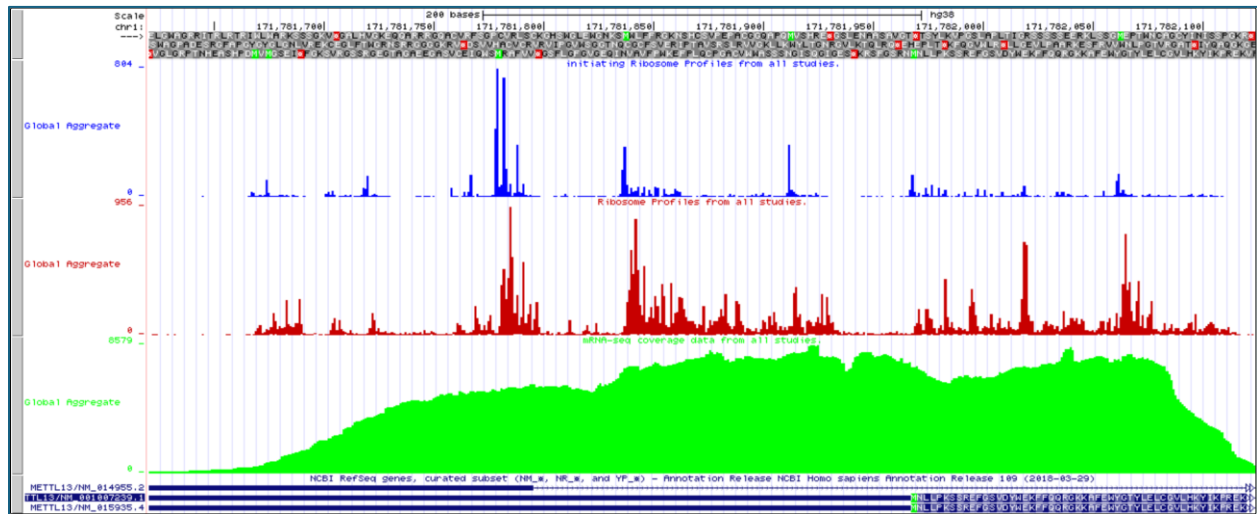
The number of SINEs and LINEs associated repeats in the mRNA: 1

Bonus question 1: Do you notice relationships between the displayed tracks? What are they?
Can you explain at least some of these relationships?

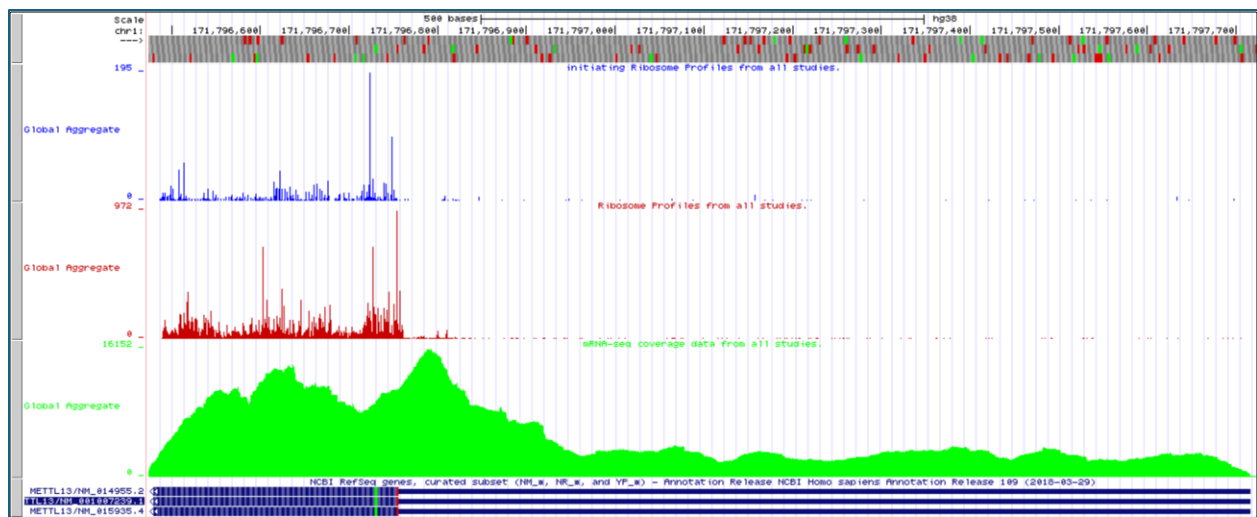
Bonus question 2: Explore your locus using other tracks available in UCSC genome browser.
If you find something interesting/unusual, describe your most interesting findings. You could illustrate it with a relevant browser screenshot. You can use only one screenshot.

2. GWIPS-viz browser.

First coding exon:

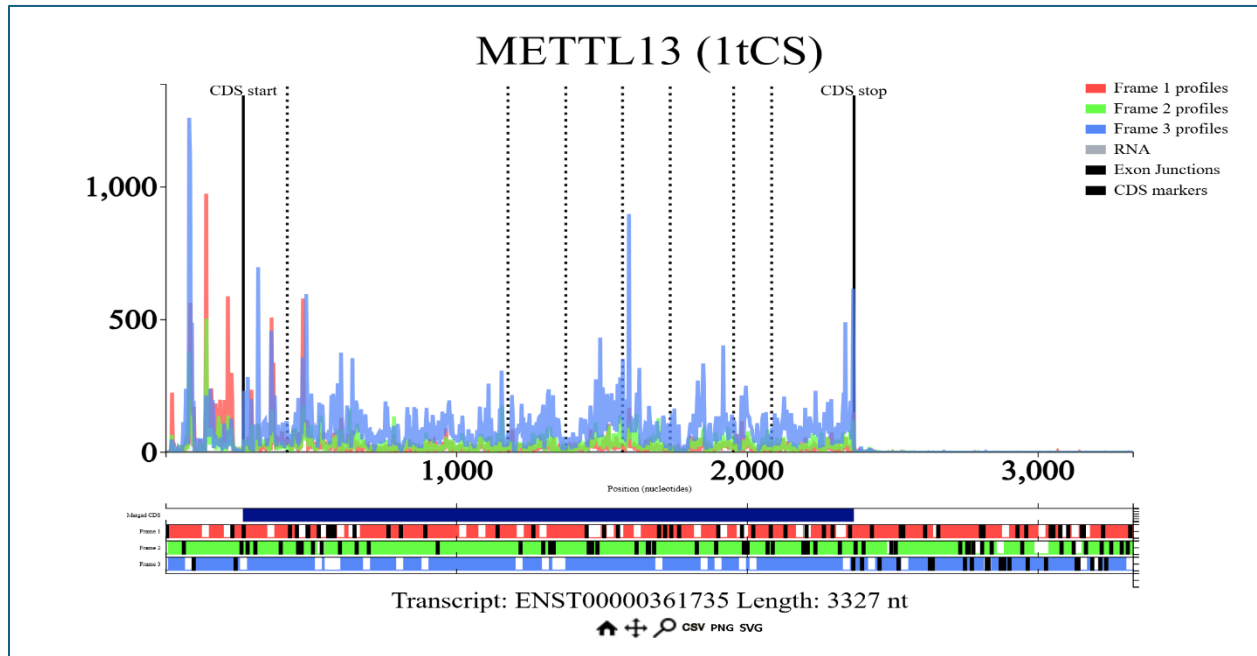


Last coding exon:



Bonus question: If you see anything unusual/unexpected, describe it.

3. TRIPS-viz browser.



Shortcode link: <https://trips.ucc.ie/short/1tCS>

Bonus question: describe all ORFs that you think are translated according to the available data.

Part V. Improvements (bonus part).

1. Suggestions.

Suggest other bioinformatics resources or topics that should become a part of this practical.

2. Justification.

Explain why do you think it is important to include these resources in this practical.

3. Should UCC provide a separate module on programming/computing/data analysis for life scientists?

Yes, I believe so because as somebody who is interested in bioinformatics but comes more from a math/statistics background than biology, a more analysis/computing heavy class would align greatly with my interests. At the moment, there isn't a class in any department that touches on programming heavy bioinformatics, and that would be very helpful for somebody like me, potentially interested in that kind of research.